Product Requirements Document

VERSION 1.1 | 06 NOVEMBER 2026

400 COFFEE

Product Requirements Document

Product definition, scope, workflows, requirements and release criteria

Document status: Professional working draft for stakeholder review. Open decisions and assumptions are explicitly identified.

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1. Document control

Purpose Define the product requirements for a multi-tenant café operations SaaS, validated through a pilot at 400 Coffee. This document is the baseline for product, design, engineering, QA and commercial planning.

Status Baseline draft for stakeholder review. Requirements marked TBD require confirmation during discovery. Version 1.0.

Product principle Build a repeatable SaaS product, not a one-off billing application. 400 Coffee is the first operational pilot and design partner.

2. Product vision and outcomes

Vision One connected operating system for independent cafés: cashier ordering, customer self-service, split kitchen execution, payment reconciliation and owner visibility.

Primary outcomes Reduce order-entry friction; reduce missed or delayed kitchen tickets; improve payment traceability; give owners a reliable view of sales and collections; make onboarding repeatable across tenants.

Success measures Order capture accuracy; order-to-kitchen latency; percentage of orders with complete station tickets; payment reconciliation exceptions; end-of-day collection variance; staff task completion time; pilot staff adoption.

3. Context and assumptions

Operating context Current operations are largely manual. Payments are digital through UPI. Expected volume is approximately 200–500 orders/day. Service includes dine-in and takeaway. A cashier-led workflow and static UPI QR are the initial decisions.

Operating model One outlet initially, with beverage and food preparation stations. Hardware selection remains flexible. Customer QR order preparation/release policy must be configurable.

Payment constraint A static UPI QR identifies the merchant, not a unique order. Customer self-declaration or a screenshot is not payment proof. During the pilot, cashier/authorized staff verify collection in the merchant-side bank/UPI channel and record the verification.

Assumptions to validate Menu, tax treatment, service charge, invoice format, table model, refund authority, UPI merchant account/API availability, printer requirements, peak concurrency, staffing roles and support hours.

4. Personas and roles

Cashier Creates and edits orders, selects dine-in/takeaway, dispatches orders, records cash or verified UPI collection, handles permitted cancellations and prints/shares bills.

Kitchen staff Works assigned beverage or food tickets, starts preparation, marks items/tickets ready and sees modifications and special instructions.

Owner/manager Monitors sales, order flow, payment exceptions, shifts, menu performance and operational configuration.

Customer Browses menu via QR, configures eligible items, submits an order, receives status updates and follows pickup/table instructions.

SaaS platform operator Onboards tenants, configures plans and entitlements, monitors platform health and provides controlled support without unrestricted tenant-data access.

5. Scope and release boundaries

Pilot / Release 1 Cashier POS; dine-in and takeaway; menu and modifiers; customer QR ordering; configurable order release policy; split beverage/food KDS; payment ledger and manual UPI verification; bills; shifts; owner dashboard; audit trail.

Release 2 Production SaaS hardening; self-service tenant onboarding; subscriptions and entitlements; tenant administration; exports; offline recovery; monitoring; payment integration where supported by provider/bank.

Release 3 Inventory and recipe foundation; stock consumption; wastage and adjustment workflows.

Release 4 Loyalty, customer engagement, advanced analytics, AI owner assistant, multi-outlet expansion and delivery integrations.

Out of scope for pilot Automatic UPI payment confirmation without an approved merchant integration; customer app installation; delivery marketplace integrations; autonomous AI financial actions; full ERP/accounting replacement.

6. Functional requirements

FR-001 Tenant and outlet The platform shall isolate tenant data and permissions. Users shall access only their authorized tenant/outlet. Platform support access shall be purpose-limited and audited.

FR-002 Catalog Authorized users shall manage categories, products, prices, availability, taxes/charges, modifiers, station routing and display order. Price and tax values shall be versioned/snapshotted on each order.

FR-003 POS order creation Cashier shall create dine-in or takeaway orders, add items/modifiers/notes, change quantities, review server-calculated totals and submit orders. The system shall assign a unique human-readable order number.

FR-004 Order lifecycle Order transitions shall be server-authoritative, validated and audited. Duplicate submissions shall be idempotent. Cancellations and edits after kitchen dispatch shall follow role-based rules.

FR-005 Customer QR Customer shall open a secure outlet/table QR experience without installing an app, browse menu, configure eligible items, submit as guest and track status using a secure token.

FR-006 QR release policy Outlet shall configure whether QR orders are held for cashier acceptance or released automatically under defined conditions. The policy shall be snapshotted when an order is submitted.

FR-007 Split kitchen A parent order shall generate station-specific tickets for beverage and food stations. Each station shall see only relevant items and modifiers. Parent readiness shall be computed from all required station tickets/items.

FR-008 KDS execution Kitchen staff shall see new tickets in real time, sort/filter queues, start preparation, mark ready and view elapsed time. Reconnect shall recover missed events from authoritative APIs.

FR-009 Payments The system shall maintain payment attempts and ledger entries separately from order state. It shall support cash, UPI declaration, manual verification, partial payment where enabled, overpayment exception, refund records and reconciliation.

FR-010 UPI verification Customer declaration shall not mark an order paid. Authorized staff shall verify merchant-side receipt, enter amount/reference and record verifier/time. Duplicate references and amount mismatches shall be blocked or flagged.

FR-011 Billing and shifts The system shall generate itemized bills/receipts, maintain cashier shifts, record opening/closing cash where applicable and produce shift reconciliation.

FR-012 Owner dashboard Dashboard shall show orders, gross order value, confirmed collections, outstanding balances, refunds, net collections, payment mix, AOV, popular products and delayed orders. Metrics shall be clearly defined.

FR-013 Notifications System shall notify relevant roles of new orders, payment exceptions, delayed tickets and status changes using in-app events first; external channels are configurable.

FR-014 Audit Critical actions shall record actor, tenant/outlet, timestamp, entity, before/after state where appropriate, reason and request correlation ID.

FR-015 Offline resilience Staff app shall preserve locally entered unsubmitted work and clearly indicate sync status. Acknowledged server orders shall not be lost. Customer QR submission shall require online server acceptance in V1.

FR-016 SaaS administration Authorized platform operators shall manage tenants, outlets, plans, entitlements, subscriptions, service status and support actions with least privilege.

7. Non-functional requirements

Performance targets Initial targets for validation: p95 ordinary reads <500 ms; p95 order submission <1 s excluding external payment services; kitchen updates typically <2 s; load test at 3× measured peak.

Availability and recovery Initial planning target: 99.5% monthly availability; RPO 15 minutes and RTO 4 hours. These are proposed targets, not contractual commitments until infrastructure and operating model are approved.

Security TLS in transit; secure password/session handling; role-based and tenant-aware authorization; secret management; rate limiting; input validation; audit logging; dependency scanning; backup encryption; least-privilege access.

Privacy Collect only necessary customer data. Phone number should be optional unless an approved operational need exists. Define purpose, retention, deletion, access and processor roles before production.

Usability Tablet-first staff workflows, large touch targets, explicit status text, high contrast, minimal typing, error recovery without losing order state.

Scalability Modular monolith initially; tenant-aware data model; asynchronous event processing where useful; avoid premature microservices.

8. Key workflows

Cashier order Select order type → add items/modifiers → review server totals → submit → station tickets created → collect/record payment → issue receipt → close/serve order.

Customer QR Scan outlet/table QR → browse → customize → submit → cashier accepts or configured auto-release → station tickets → status tracking → pickup/serve.

Split kitchen Order accepted → partition items by station → beverage and food tickets progress independently → parent ready only when all required tickets are ready.

UPI payment Create amount-bound payment attempt → show static merchant QR and reference → customer declares → cashier checks merchant-side receipt → enter verified amount/reference → ledger records verification → order payment status updates.

Shift close Review sales and payment ledger → reconcile cash and UPI → investigate exceptions → record counted values and variance → close shift with audit record.

9. Acceptance and launch gates

Data integrity No lost acknowledged orders; no duplicate fulfillment; totals recomputed server-side; every payment movement ledgered; no false paid status.

Kitchen reliability Station tickets recover after reconnect; parent order state accurately aggregates split tickets; cancellation/edit changes are propagated and audited.

Security Automated tenant isolation tests pass for APIs, exports, events and support tools. No cross-tenant access.

Operational readiness Peak-hour simulation passes; staff complete common tasks; devices remain usable during a full service window; recovery and support runbooks approved.

Commercial readiness Pricing, subscription terms, tax treatment, support scope, data terms and pilot acceptance are approved by the contracting parties.

10. Open decisions

Business Confirm legal names and addresses of supplier and customer; outlet/table model; menu/tax rules; bill numbering; refund/cancellation policy; service hours; support SLA.

Technology Confirm hosting region, deployment environment, identity provider, observability stack, printer model, tablet model and backup/restore procedures.

Payments Identify UPI merchant provider/bank and available APIs/webhooks; define evidence acceptable for manual verification; define reconciliation and refund permissions.

Commercial Confirm whether 400 Coffee is a paid pilot, design partner, or internal demonstration; confirm ownership/licensing of custom features and data.

VERSION 1.1 — CUSTOMER EXPERIENCE ADDENDUM

Version 1.1 addendum — 06 November 2026. This addendum supersedes any earlier customer-facing UX description that implied a conventional QR menu/dashboard as the primary customer experience. The immersive Virtual Café is the approved customer-side direction. Existing staff POS, KDS, owner and platform-admin interfaces remain separate workstreams and are not redesigned by this customer-side change.

1.1 Approved customer experience direction

The customer-facing experience is “400 Coffee — The Virtual Café”: an immersive, cinematic, coffee-and-food themed ordering journey. It is not a conventional dashboard, generic e-commerce storefront or plain QR menu. The first customer-side build must prioritize atmosphere, storytelling and a premium, low-friction ordering flow. The selected direction is: interactive experience; 3D manager character; cinematic camera/scene transitions. This is a customer-side scope only. Do not redesign or build the restaurant admin/POS/KDS/owner interfaces as part of this UI increment.

1.2 Customer journey — required end-to-end

1. Entry: customer opens the café experience from a QR code/app/PWA entry point. Show a cinematic camera move into a warm café: walnut counter, espresso machine, shelves, warm amber practical lighting, subtle steam/particles and layered depth. The manager is visible behind the counter and greets the customer: “Welcome to 400 Coffee. How can I help you today?” Provide two clear choices: “View menu & place order” and “Pay bill”. The scene must load quickly and degrade gracefully if WebGL/3D is unavailable.

2. Menu: choosing the menu opens a vintage, leather-bound café diary/menu with page-turn or equivalent tactile transition. Show menu categories, product names, descriptions, actual configured prices, availability and genuine product photography/approved imagery. Categories include Coffee, Hot Beverages, Cold Beverages, Fresh Juices, Sandwiches, Snacks and Desserts; final categories and products are merchant-configurable. Never hardcode sample prices into production.

3. Product customization: customer selects a product, quantity, size/variant, allowed add-ons and preferences. Provide free-text preparation/cooking instructions where relevant (e.g. less spicy, extra toasted, no ice). Clearly show price deltas and updated line total. Respect product availability and modifier constraints from the server.

4. Basket: show items, selected variants, add-ons, instructions, quantities, remove/edit controls, item totals, taxes/charges/discounts if configured, and a server-authoritative grand total. Preserve basket through navigation and recover from refresh where the session allows.

5. Identity: manager asks for customer name and mobile number. Guest checkout is the default; do not force account creation. Validate Indian mobile format on the client and server. Explain why the phone number is requested and obtain required consent for transactional WhatsApp messages. Avoid marketing consent being bundled with transactional consent.

6. Review: customer sees a complete final order summary, contact details, fulfilment context (takeaway/dine-in if enabled), all price components and total. Customer explicitly confirms the order. Back/edit must preserve entered information.

7. Payment choice: manager asks whether to “Pay now” or “Pay at takeaway counter”. Pay-later submits an order with a clearly labelled payment-due state; it is not marked paid. Pay-now offers UPI, QR, credit card and debit card only when configured and supported by the provider.

8. Payment: create a payment intent/session server-side. Display provider-generated dynamic QR/deep link/hosted checkout/terminal flow as applicable. A static illustrative QR must never be presented as a verified order-specific payment. Client callback, return URL, screenshot or user declaration is not sufficient to mark paid. Confirm payment only from a verified server-side provider webhook/status API or authorized staff verification under the pilot's manual UPI process.

9. Confirmation: after successful order submission, show a unique order reference, line items, total, fulfilment/payment state and next-step instructions. For pay-later, explicitly state payment is due at the takeaway counter. For pay-now, show paid only after verified payment. Send WhatsApp order/payment confirmation only after backend order creation and the appropriate consent. Do

not claim a WhatsApp message was sent if the provider call failed or is not connected.

10. Restaurant sync: accepted customer order is created in the shared backend and appears in the restaurant-side order workflow according to the configured cashier-acceptance/release policy. The customer UI must show “submitted / awaiting café acceptance” versus “accepted / preparing” distinctly. The restaurant-side application is not part of this customer-side UI build, but the integration contract must be implemented and tested against a mock or real backend.

1.3 Product requirements additions

FR-CX-001 Cinematic entry: deliver an interactive café scene with camera entrance, manager greeting, ambient motion and two primary actions. Respect reduced-motion preference and provide accessible skip/continue controls.

FR-CX-002 Graceful 3D fallback: if WebGL, GPU, low-memory device or network prevents the scene, render a high-quality 2D café fallback with identical functionality.

FR-CX-003 Character interaction: manager dialogue is state-driven, contextual and never blocks essential controls. Do not simulate an AI agent unless an actual approved assistant is integrated.

FR-CX-004 Diary menu: category navigation and page transitions must preserve category, scroll and basket state; support keyboard and touch.

FR-CX-005 Catalog integrity: menu, price, modifier, availability, tax and routing values come from backend APIs; the server recalculates all totals.

FR-CX-006 Basket and instructions: support quantity, variants, allowed modifiers, preparation notes, edit/remove and server validation.

FR-CX-007 Guest identity: collect only required name/mobile fields; validate and obtain transactional notification consent as required.

FR-CX-008 Review/confirmation: require explicit final confirmation; protect against duplicate taps with idempotency keys and clear loading/error states.

FR-CX-009 Payment selection: show only enabled methods. Payment attempt and order state remain separate. Payment state is server-authoritative.

FR-CX-010 Pay at counter: create order as unpaid/due, and show payment due at takeaway. Do not hold the customer on the payment screen.

FR-CX-011 Notifications: queue WhatsApp messages through an approved provider after order/payment events; persist delivery status and retry safely.

FR-CX-012 Restaurant handoff: publish order-created/accepted/status events; recover missed updates through API polling/reconnect.

FR-CX-013 Bill payment: allow lookup by order/bill reference and phone verification, then present only the outstanding balance. Do not accept a customer-entered arbitrary amount in production.

FR-CX-014 Accessibility/performance: responsive mobile-first design, screen-reader labels, keyboard operation, sufficient contrast, reduced motion, loading skeletons and 2D fallback.

1.4 Scope and non-goals

Included in the customer-side increment: responsive PWA/web experience, cinematic scene, manager character, diary menu, product customization, basket, guest identity, review, pay-now/pay-at-counter choice, configured payment handoff, confirmation, bill lookup/payment entry, customer order status and WhatsApp integration hooks. Excluded: restaurant staff POS/KDS/owner redesign, inventory/recipes, loyalty, delivery marketplace, autonomous AI ordering, actual card acquiring without provider credentials, automatic static-QR reconciliation, and production messaging without approved WhatsApp Business setup.

1.5 Customer experience acceptance gates

A first-time customer can reach menu and add an item without account creation. All menu prices and totals match server calculation. Customizations and preparation notes persist into the order and station tickets. Customer can edit before final confirmation. Repeated taps/network retries cannot create duplicate orders. Pay-at-counter orders are never marked paid. Payment success is only reflected after server verification. Confirmation contains the same order reference and amount as backend. WhatsApp failure does not lose the order. 3D failure does not prevent ordering. Reduced-motion mode avoids forced camera movement. UI works on common Android/iOS mobile browsers and desktop browsers.

1.6 Content and menu data governance

Any menu in design prototypes is illustrative only. The café owner must approve product names, descriptions, prices, photos, sizes, add-ons, allergens, vegetarian/non-vegetarian markers, availability windows, taxes and preparation notes before production. No dietary/allergen claims may be invented. Images must be licensed/merchant supplied. Menu content is data, not source code.

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